

Cell Cycle & Cell Division

Top 20 Questions (100% from this pdf In NEET)

Q1 Correct statement

- a) prophase of mitosis long, complex than prophase of meiosis-I
- b) prophase-I subdivided into 5 phase (criteria: centromere behaviour)
- c) Meiosis reduce chromosome number half, formed diploid cell
- d) meiosis ensure haploid phase in sexually Reproducing organism & Fertilisation Restore diploid phase

(D)

Q2 Match the following

- | | |
|--------------|---|
| 1 Meiosis-I | a) Reductional div ⁿ , occur in 2n |
| 2 Meiosis-II | b) equational div ⁿ , occur in n |
| 3 Leptotene | c) crossing over, / Recombination |
| 4 Zygotene | d) chromosome visible in Light microscope |
| 5 Pachytene | e) synapsis / synaptonemal complex |
| 6 Diplotene | f) Terminalisation of chiasmata |
| 7 Diakinesis | g) chiasmata |

1-a, 2-b, 3-d, 4-e, 5-c, 6-g, 7-f

Q3 Incorrect

- a) pairing of homologous chromosome: synapsis
- b) Recombination nodule is site where crossing over takes place between non-sister chromatid of homologous chromosome: pachytene
- c) Bivalent / Tetrad visible at pachytene
- d) Recombinase in pachytene
- e) None

(e)

Q4 Incorrect

- a) dissolution of synaptonemal complex begins & homologous chromosome begins to separate in diplotene
- b) chiasmata where crossing over occurred (No separation) - X shape
- c) compaction of chromosome continue in Leptotene
- d) Diplotene is Transition to metaphase-I

(d)

Q5 Correct statement

- a) Final stage of meiotic-I is diakinesis
- b) Leptotene $\rightarrow$ zygotene $\rightarrow$ Diplotene $\rightarrow$ pachytene $\rightarrow$ Diakinesis
- c) chromosome fully condensed & meiotic spindle is prepare for Seperation of homologous chromosome in diakinesis
- d) By early diakinesis nucleolus, nuclear membrane disappear
- e) Univalent chromosome align on equatorial plate in metaphase-I
- f) One metaphasic plate in metaphase-I

a, c,

Q6 If 20 chromosome present in parent cell, DNA is 40 pg

$G_1 \rightarrow 20, 40, 2n, 2c$

$S \rightarrow 20, 80, 2n, 4c$

$G_2 \rightarrow 20, 80, 2n, 4c$

MEIOSIS-I $\rightarrow 10, 40, n, 2c$

MEIOSIS-II $\rightarrow 10, 20, n, 1c$
(gametes)

Q7 Correct statement

- a) homologous chromosome seperate, sister chromatid not seperated: anaphase-I
- b) Nucleolus, NM disappears in Telophase-I
- c) chromosome do not reach extended state of interphase nucleus.
- d) stage b/w Two mitosis is interkinesis

(C)

Q8 Correct statement

- a) Interkinesis is short lived, Centriole & DNA duplication occur
- b) Interkinesis is followed by prophase-II, Complex than prophase-I
- c) meiosis-II Resemble mitosis
- d) chromosome compact & NM, Nucleolus reappear in prophase-II

(C)

Q9 Incorrect

- a) univalent chromosome arrange at equator, having Two metaphasic plate, spindle attach to Kinetochore of sister chromatid : (Metaphase-II) a
- b) splitting of centromere in anaphase-II (shortening of microtubule) : Anaphase-II
- c) Meiosis end with telophase-II, formation of tetrad (4 haploid cells)
- d) All four cells are disimilar to each other & to parent also

Q10 Significance of meiosis

- a) conservation of specific chromosome across generation (paradoxically) a
- b) decrease genetic variability
- c) No role in variation
- d) None

★ सारे DIAGRAM देख लेना !

Q11 Match the following

1-b, 2-a, 3-d, 4-c, 5-h, 6-e, 7-f, 8-g

- | | |
|---------------------------------------|--|
| 1 Interphase (>95%) | a) Dividing phase |
| 2 M phase (<5%) | b) cell is preparing for division (Resting) |
| 3 G ₁ (gap-1) | c) DNA Replication, centriole duplication |
| 4 S (synthesis) | d) B/w mitosis & initiation of DNA Replication |
| 5 G ₂ (gap-2) | e) yeast |
| 6 90 minutes | f) human |
| 7 23 hour (Interphase)+1 hr (M phase) | g) M phase |
| 8 Karyokinesis & cytokinesis | h) synthesis of protein for mitosis & cell growth continue |

Q12 Correct statement

- a) cell is metabolically active, does not replicate DNA : S b,d
- b) $2n \rightarrow 4C$ & $2n \rightarrow 2n$ after S phase
- c) cell division, DNA Replication, cell growth Not in coordination; cell duplicate its genome, synthesis other constituents and divide
- d) into Two daughter cell : cell cycle

Q13 parent cell : $2n$ chromosome , DNA : $2C$

G_1 : $2n$, $2C$

S : $2n$, $4C$

G_2 : $2n$, $4C$

After M phase : $2n$, $2C$ so equational division

Q14 correct

- a) Some cell (heart cell) do not divide exit G_1 phase & enter in G_0
- b) cells is active and do not divide in G_0 (quiescent stage / active phase)
- c) if cell divide occasionally to Replace lost cell due to injury so it will leave G_0
- d) cell no longer proliferate unless they are called on to do so.
- e) Both ① & ④

e

Q15 correct

- a) Mitosis restricted to diploid cell in animal (without exception)
- b) male honey bee is haploid undergoes mitosis , having 32 chromosome
- c) M phase also known as progressive / Reorganization phase
- d) All

c

Q16

- | | |
|--|----------------------------------|
| ① centrosome moves opposite pole | : Metaphase |
| ② Two aster with spindle fibre | : Mitotic apparatus |
| ③ Initiation of condensation of chromosomal material / untrangel | : Anaphase |
| ④ prophase-II | : GB, ER, Nucleolus, NM Reappear |
| ⑤ Splitting of centromere | : Metaphase |
| ⑥ Study of chromosome | : Anaphase |

②

Q17 Correct statement (Anaphase)

- a) chromosome split, daughter chromatid move opposite pole
- b) centromere : Trailing Behind & arms : lead
- c) if 4 chromosome in metaphase, then 8 in anaphase
- d) Both (a) & (b)

d

Q18 Correct statement (Telophase)

- a) final stage of cytokinesis
- b) chromosome decondense, maintain identity
- c) In G_1 , one chromosome have Two chromatid
- d) Nucleolus, NM disappear
- e) None

e

Q19 Correct statement

- a) karyokinesis is not-followed by cytokinesis so multinucleated condition arise called syncytium (liquid endosperm in coconut)
- b) cell plate method in animal & cleavage furrow in plant
- c) Mitosis only segregation of chromosome But Not division of cytoplasm.
- d) appearance of furrow in cell wall of animal cell

a

Q20 Incorrect statement (Metaphase)

- a) condensation incomplete, complete disintegration of nuclear envelope makes this phase
- b) small disc / protein over centromere : kinetochore help in attachment of spindle fibre
- c) chromosome comes at equator / metaphasic plate
- d) alignment of chromosome from plate
- e) b / c / d

a

Significance of mitosis Read कर लेना !